

ARCHITECTURAL RESEARCH METHODS

SECOND EDITION

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Chapter 12

Case Studies and Combined Strategies

12.1 INTRODUCTION

In 1961, Jane Jacobs wrote her classic book, *The Death and Life of Great American Cities.* Jacobs's book challenged the conventional wisdom of Modernist-inspired urban renewal popular at that time. Her insights about how to maintain and foster the vitality of cities are derived almost entirely from vignettes of life in New York City (see Figure 12.1).

However, the richness and depth of her many examples of the socio-physical dynamics of life in New York were powerfully persuasive; as a consequence, the book had an enormous impact on the planning and architecture professions. Moreover, the themes she identified were observed and documented in other cities in subsequent analyses. In her introduction, she presents an articulate rationale for the strategy of her investigation:

> In setting forth different principles, I shall mainly be writing about common, ordinary things. . . . The way to get at what goes on in the seemingly mysterious and perverse behavior of cities is, I think, to look closely, and with as little expectation as is possible, at the most ordinary scenes and events, and attempt to see what they mean and whether any threads of principle emerge among them. . . . I use a preponderance of examples from New York because that is where I live. But most of the basic ideas in this book come from things I first noticed or was told in other cities. . . . I hope any reader of this book will constantly and skeptically test what I say against his own knowledge of cities and their behavior.[1]



Figure 12.1 Cityscape in Brooklyn Heights, New York. Photograph courtesy of Linda Groat.

In a study of popular Modernism in Brazil, Fernando Lara poses the questions: Why was Modernist architecture better received in Brazil than in Europe or the United States? How were the attributes of Modernism promulgated such that many working and middle-class houses of the 1950s were built with visible attributes of Modernism? And to what extent were these houses truly Modernist?[2]

To explore these questions, Lara identified the Brazilian city Belo Horizonte as the focus of his study. Belo Horizonte was selected for at least two reasons: (1) the great number of 1950s popular Modernist houses in major sections of the city; and (2) the presence of one of the first and most significant ensembles of Modernist public buildings, the Pampulha complex built in the early 1940s (see Figures 12.2 and 12.3). In this regard, Pampulha represents an officially sanctioned and close-at-hand example of Modernism, known to the entire city's population.

In framing the contours of the study, Lara identified three types of data sources: (1) archival research to ascertain the influence of various social, economic, cultural,



Figure 12.2 A typical popular Modernist house, Belo Horizonte, Brazil. Photograph courtesy of Fernando Lara.

and media transmissions; (2) formal analyses (of facade design and interior layouts) in a sample of 300 houses in two sections of the city; and (3) in-depth interviews with 20–30 residents of popular Modernist houses, many of whom—though elderly—were original owners. In this way, Lara has combined multiple data sources that address the broad cultural influences, the physical extent of Modernist adoption and/or adaptation, and the residents' own understanding of their homes in relation to the larger cultural context.



Figure 12.3 Part of the Pampulha complex, Belo Horizonte, Brazil. Photograph courtesy of Fernando Lara.

From this rich array of data sources, Lara is able to explain in very detailed and nuanced ways how and why Modernism came to infuse the building of middle-class residential areas of Belo Horizonte. Indeed, some of his analyses uncover unexpected patterns and relationships. For example, although popular magazines were full of both advertisements and stories about the artifacts of modernization, many of the families who built Modernist-style homes were most influenced through personal connections and/or direct exposure to the homes of more well-to-do families (see Figure 12.4).

Moreover, the application of Modernist principles was often inconsistent or piecemeal. For example, Modernist features were applied to the facade of houses, while the interior layouts reflected the social traditions of pre-Modernist houses (see Figure 12.5).

These two studies illustrate two powerful, and sometimes overlapping, approaches to research design. Jacobs's study is a preeminent and well-respected example of the *case study* strategy; she uses the example of New York City—as a particular case—to explore the multiple socio-physical dynamics that contribute to the vitality of urban life. Although she may have gained insights from her experience in other cities, and others may have studied other cities in light of her conclusions, the heart and soul of her study, is about the particular case of New York City.

Lara's research is also a case study, in that he focuses on the multifaceted dynamics that led to adoption and adaptation of Modernism in one city: Belo Horizonte, Brazil. But his study also represents quite emphatically the power of *combined strategies,* in this instance the historical and the qualitative strategies.

In the remainder of this chapter, we will examine in detail both the case study strategy and models for achieving effective *combined strategies.*

12.2 STRATEGY: GENERAL CHARACTERISTICS OF THE CASE STUDY

In the one of the most frequently cited books on case study research, Robert Yin provides the following definition: "A case study is an empirical inquiry that investigates a contemporary phenomenon within its real-life context, especially when the boundaries between phenomenon and context are not clearly evident."[3] To make the definition more clearly applicable to architectural research, we would amend Yin's definition to read: an empirical inquiry that investigates a phenomenon or setting. By deleting the word *contemporary* and adding the word *setting,* this definition would specifically accommodate the explicit inclusion of historic phenomena and both historic and contemporary settings as potential foci of case studies.

What, then, are the primary identifying characteristics of the case study? Briefly, the five particularly salient characteristics are: (1) a focus on either single or



Figure 12.4 A typical product advertisement, Brazil, 1950s. Courtesy of Fernando Lara.

multiple cases, studied in their real-life contexts; (2) the capacity to explain causal links; (3) the importance of theory development in the research design phase; (4) a reliance on multiple sources of evidence, with data converging in a triangular fashion; and (5) the power to generalize to theory. These five characteristics will be discussed in detail in the following chapter segments.

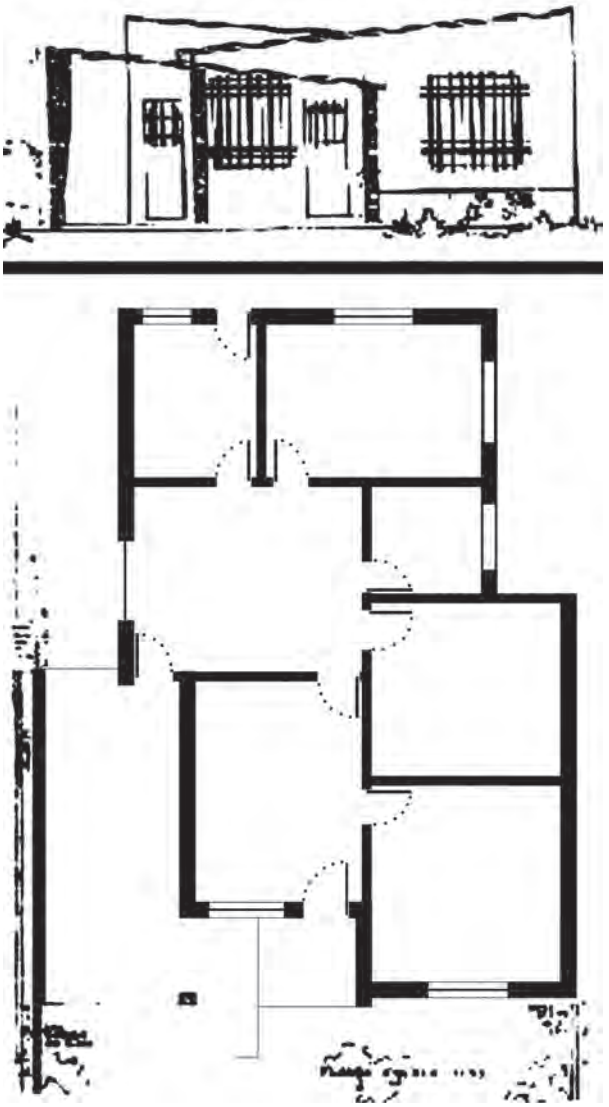


Figure 12.5 Typical facade and floor plan from Lara, 2008. Courtesy of Fernando Lara.

12.2.1 A Focus on Cases in Their Contexts

The essence of the case study strategy is its focus on studying a setting or phenomenon embedded in its real-life context. As Yin describes it, the case study strategy implies much more than simply studying a phenomenon in the "field." Rather, the case study involves studying a case in relation to the complex dynamics with which it intersects and from which the case itself is inseparable. This definition of the "case" is clearly evident in both the Jacobs and the Lara studies. For instance, Jacobs's investigation of urban vitality in the case of New York City is substantially linked to a multitude of contextual factors and phenomena—from the rise of the automobile culture, to federal funding policies, to trends in planning theory. Similarly, Lara's study of Modernist houses in the case of Belo Horizonte entails a wide range of issues from the role of modernization in Brazil, to the economic prosperity of the 1950s, to the influence of local political leaders. As both of these examples demonstrate, the context of the case becomes virtually inseparable from the definition of the case itself.

BOX 12.1
Case Study: A 100% Flexible Workspace

A combination of recent trends—including the globalization of markets, rapid growth of telecommuting and telecommunications, the shift toward collaborative work or project teams, and the increasingly flexible assignment of office space, such as hot desks—has led many organizations to reconsider how they plan for and use their office environments. In some organizations, the changes have been so profound that it is unclear how, or if, ongoing work practices are being supported by the design of the physical environment. In particular, Janice Barnes wanted to investigate this question: How does the design of the physical environment support the way project teams share knowledge in a 100% flexible workplace?

The focus of this case study research was one office site within a major global consulting organization of over 130,000 employees, with operations in over 50 countries. The particular site was chosen because it had recently been designed by a local architectural firm to meet the new workplace standards established by the organization in 2000. The design intention was to provide a 100% flexible workplace in which no person, even at the partner level, claims a dedicated office. All office assignments are temporary, and extensive telecommuting is accommodated.[a] In addition to site

[a] Janice Barnes, *Situated Cognition in Flexible Work Arrangements.* PhD dissertation, University of Michigan, Ann Arbor, 2001.

(*Continued*)

visits for observations, Barnes conducted open-ended interviews with approximately 25 people, including members of various project teams, office support staff, facilities department staff members, human resources staff, a national facilities staff member, and several members of the architectural firm that had designed the site.

Barnes concludes that in this new environment, project teams most frequently share knowledge via an array of artifacts displayed in the team work room (see Figure 12.6). These artifacts may include reference documents from clients, workflow diagrams pinned to wall surfaces, diagrams from brainstorming sessions, and so on. Few, if any, representations of knowledge are located or displayed in individual offices or cubicles. And since these offices are constantly reassigned, virtually no personalization of offices occurs. This new design standard for the workplace seems to serve the short-term interests of the project teams reasonably well, but interviews revealed that employees felt little, if any, attachment to the workplace and invested no energy in getting to know other employees within it. Much as artifacts exhibit short life spans in the organization (duration of the project), so too do many working relationships.



Figure 12.6 Typical project team work room, with the artifacts representing the project team's shared knowledge. Drawn by Fernando Lara.

12.2.2 The Capacity to Explain Causal Links

One of the most frequently discussed issues in research design is that of causality. As we discussed in Chapter 9, for instance, the experimental research design is fundamentally orchestrated so as to ascertain the causal capacity of the independent or treatment variable. By contrast, correlational design can identify patterns of relationships, but stops short of attributing cause (see Chapter 8). Yet we have also seen that both interpretive-historical and qualitative strategies can also address the issue of causality, albeit in quite a different way than experimental research; both of these strategies offer the potential to uncover the multiple, complex, and sometimes overlapping factors that eventually lead to particular outcomes. It is in this latter sense that case studies can also identify causal links among an array of socio-physical factors and events.

In arguing that case studies can, like experiments, be *explanatory*, Yin suggests that case studies can also be either *descriptive* or *exploratory* in purpose.[4] Whether a particular case study is explanatory, descriptive, exploratory, or some combination of these is a function of the researcher's purpose—or more precisely the nature of the research question—rather than any limitation inherent in the case study strategy. To clarify his point, Yin develops a case study typology that distinguishes among both research goals and design structure.

	Purpose of Case Study		
Type of Structure	**Explanatory**	**Descriptive**	**Exploratory**
1. Linear-Analytic Typical article format: problem statement literature review methods results	X	X	X
2. Chronological (narrative sequence)	X	X	X
3. Theory-Building Sequence of chapters depends on logic of theory development	X		X
4. Unsequenced Sequence of chapters interchangeable		X	

Figure 12.7 Typology of case study designs. Adapted from Robert K. Yin, *Case Study Research: Design and Methods* (Thousand Oaks, CA: Sage, 1994), 138. (Yin includes two other types not commonly employed in architectural research.) Courtesy of Robert K. Yin and SAGE Publications.

Using Yin's typology, we can classify Jacobs's study as a theory-building study that has both exploratory and explanatory purposes. The exploratory nature of her investigation is reflected in the following sentiment, quoted earlier in greater length: "The way to get at what goes on in the . . . behavior of cities is, I think, to look closely . . . and attempt to see what they mean and whether any threads of principle emerge among them."[5] But Jacobs also clearly intends to do more than just find out; she also wants "to *explain* [emphasis ours] the underlying order of cities."[6] To this end, she identifies what she calls the most "ubiquitous principle" right off the bat in her introduction:"[T]he need of cities for a most intricate and close-grained diversity of uses that give each other constant mutual support, both economically and socially."[7] This combination of exploratory and explanatory goals is clearly articulated in the four-part structure of her book. In Part One, the exploratory component is represented in her observations on the nature of cities. In Part Two, Jacobs lays out the heart of her explanatory argument by identifying four key conditions for city diversity. Finally, the next two sections discuss the implications of the diversity principle for the regeneration of cities, including specific tactics for achieving such regeneration.

Similarly, Lara's study of Brazilian Modernism is driven primarily by an explanatory purpose. His basic research questions are these: Why and how was Modernism so much more enthusiastically embraced by the Brazilian middle class than it was in the United States and most of Europe? Secondarily, Lara's research also reveals both exploratory and descriptive purposes. With respect to the former, he seeks to explore the complex dynamics of Modernism's infusion into Brazilian culture through multiple—and previously unexamined—materials, including archives, documentation of the physical artifacts, and oral histories. With respect to the latter, Lara's exhaustive mapping of Modernist houses in two multiblock neighborhoods of Belo Horizonte is a descriptive feat of major proportions. In the light of Yin's typology, Lara's study clearly represents the *linear-analytic* type of case study, a structure that follows the traditional outline for an academic research study: identification of a research question, literature review, methods, findings, discussion, and conclusions (see Figure 12.7). Not only does this organizing structure accommodate Lara's multiple purposes, but it is also the most conventionally suitable for a scholarly book.

12.2.3 The Role of Theory Development

Despite the relatively open-ended and broad qualities of the case study focus, described in the previous section, Yin recommends that the case study research design

BOX 12.2

Case Study: The Public Design Process for the Seattle Public Library

The Seattle Public Library, designed by world-renowned architect Rem Koolhaas, is considered by many in the architectural and design communities to be one of the most notable built projects in recent years (see Figures 12.8 and 12.9). While most architectural critics and scholars have lavished considerable praise on the building's innovative design concept, Sharon Mattern's case study of the project focuses instead on the lengthy and complex process of public input over the years both preceding and during Koolhaas's role in the project.[a]



Figure 12.8 Exterior of Seattle Public Library by Rem Koolhaas. Photograph courtesy of David Wang.

[a] Sharon Mattern, "Just How Public Is the Seattle Public Library?" *Journal of Architectural Education* 57(1) (2003): 5–18.

(*Continued*)



Figure 12.9 Interior of Seattle Public Library atrium by Rem Koolhaas. Photograph courtesy of David Wang.

To illuminate the role that the various stakeholders played in the funding, planning, and debate on the project, Mattern relies on a broad array of source material, including public documents, memos, newspaper articles and opinion pieces, presentation materials, and personal interviews. On one level Mattern's article clearly follows the historical narrative of how the library administrators and the architect sought to engage the public throughout the process. Simultaneously, Mattern illuminates the frustration that arose among the key actors concerning the role they envisioned for themselves in the design process.

Mattern concludes that right from the start the public was given the impression that they would play a major role in the physical planning and design of the building by virtue of both their financial support (through their vote for funding) and the grassroots public meetings held by the city librarian. But, as Mattern argues: "[R]ight from the beginning of the process, the library and design team delimited the field of imaginable ideas and framed the discourse surrounding the project."

Mattern also extends her analysis beyond the confines of the project itself, and the particular personalities involved, to suggest the implications for a broader understanding of engagement in the public realm: "The communications within and around a public design project serve not only in deliberating over the design itself but also in negotiating just how "public" a public space will be. . . . [A]rchitects are in the business of building not only buildings, but also consensus—and, in the process, values, identities, and ideologies."

be guided by theoretical development. As he puts it: "[T]heory development as part of the [research] design phase is essential, whether the case study's purpose is to develop or to test theory. . . . The complete research design [should embody] a 'theory' of what is being studied."[8] He then goes on to explain that by *theory*, he does not mean a "grand" theory; rather, the goal is to have "a sufficient blueprint for your study"[9] that will suggest what data must be collected and what criteria should be used for analyzing it. Perhaps it is fair to say that the role of theory development Yin proposes has some equivalent to the notion of "hypothesis" in much postpositivist research.

In both Jacobs's and Lara's studies, the role of theory development in the research design is evident. As Jacobs reveals in her introduction:

> [M]ost of the basic ideas in this book come from things I first noticed or was told in other cities. For example, my first inkling about the powerful effects of certain kinds of functional mixtures in the city came from Pittsburgh, my first speculations about street safety from Philadelphia and Baltimore, my first notions about the meanderings of downtown from Boston, my first clues to the unmaking of slums from Chicago.[10]

These observations from other similar cities prompted the theoretical propositions underpinning her case study research in New York, as she put it, "at my own front door."[11]

However, it was Lara's observations of *dissimilarity* between 1950s middle-class housing in Brazil and the United States that prompted the theory development underlying his case of Belo Horizonte. Thus, Lara's research design was guided by his intention to *explain how* and *why* popular acceptance of Modernism was so much more pervasive in Brazil than in more developed industrialized countries in Europe or in the United States.[12]

12.2.4 Using Multiple Sources of Evidence

Another key feature of the case study is its incorporation of multiple sources of evidence. Thus, while Jacobs, as a resident and participant observer of her Greenwich Village neighborhood, focuses her attention and analytical insights on the case of New York City, she also draws heavily and freely on her observations of other cities, as well as from commentaries of officials and community leaders in various cities.

Lara's case study is particularly notable for the range and variety of data sources, which included archives, oral history, and artifactual inventories, as well as formal and spatial analyses. To be specific, the archival work included (1) an examination of all issues of popular housing magazines from 1950 through 1959, which yielded hundreds of pages of articles about, and advertisements of, Modernist housing and building products; and (2) a review of 25 house plans filed with the Buildings Office of Belo Horizonte (see Figure 12.10). In addition, Lara conducted 21 interviews with family members, most often elderly widows, still living in their Modernist-inspired houses of the 1950s. These interviews were particularly



Figure 12.10 House plan diagrams. Courtesy of Fernando Lara.

important in uncovering the informal relationships and experiences that influenced the residents' choice of housing style. Third, Lara and his research team conducted a thorough artifactual survey of all houses within two multiblock neighborhoods to ascertain the proportion of Modernist-inspired housing. Finally, Lara analyzed: (1) a sample of houses in order to classify these houses along a continuum of traditional to Modernist, based on their use of specific facade design features; and (2) conducted both a visual and computer-based analysis of the spatial qualities of the floor plans (see Figures 12.11 and 12.12).[13]

12.2.5 Generalizability to Theory

Although a conventional criticism of case study research is that there is no basis for generalizing from one case to other cases, Yin contests this argument very vigorously. In effect, he argues that the premise of much correlational research—that one can

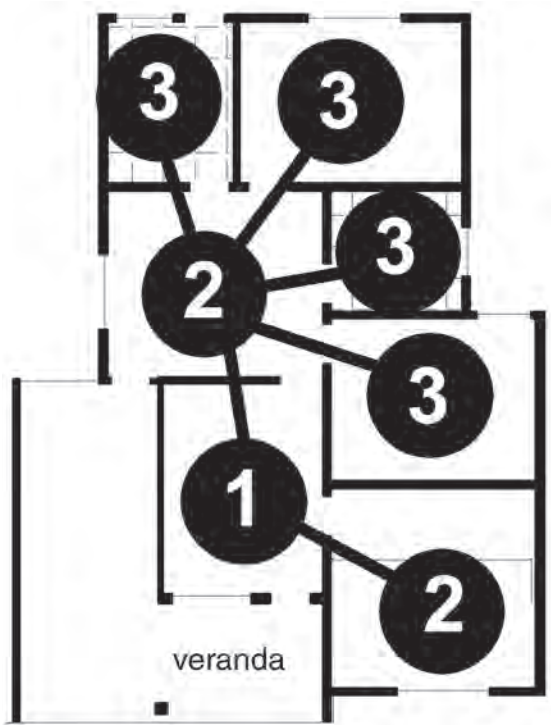


Figure 12.11 Depth analysis of a traditional house. Courtesy of Fernando Lara.

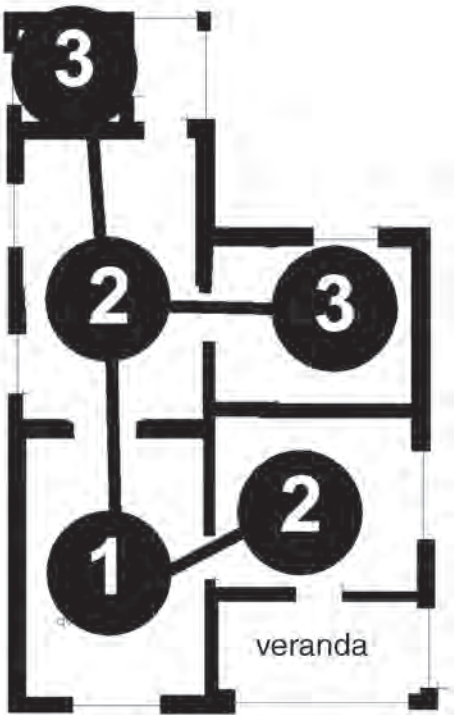


Figure 12.12 Depth analysis of a Modernist house. Courtesy of Fernando Lara.

only generalize from a representative sample to a larger population—is beside the point. Instead, he maintains, the case study's strength is its capacity to generalize to theory, much the way a single "experiment" can be generalized to theory, which can in turn be tested through other experiments. To substantiate his point about generalizability, Yin actually cites Jacobs's study. His insights are worth quoting in some detail:

> The book is based mostly on her experiences from New York City. However, the chapter topics, rather than reflecting the single experience of New York, cover broader theoretical issues in urban planning, such as the role of sidewalks, the role of neighborhood parks, the need for primary mixed uses, the need for small blocks, and the processes of slumming and unslumming. In the aggregate, these issues in fact represent the building of a theory of urban planning.

Jacobs's book created heated controversy in the planning profession. The result, in part, was that new empirical inquiries were made in other locales to examine one or another facet of her rich and provocative ideas. Her *theory*, in essence, became the vehicle for examining other cases, and the theory still stands as a significant contribution to the field of urban planning.[14] In other words, the influence and power of Jacobs's case study lies in the robust theory building she was able to achieve.

In a similar vein, the potential power of Lara's research is in the theoretical principles he is able to identify concerning the complex dynamics of modernization, political influence, indigenous architectural trends, and so on. The implication of his case study is not that the acceptance or rejection of Modernist architecture can be predicted and thereby generalized to other contexts; rather, its power will be to clarify how and why particular cultural factors can affect middle-class receptivity to architectural trends.

In contrast to Yin's stance on theory building, some advocates of the case study warn researchers that too great a focus on generalizing to theory can obscure the intrinsic value and uniqueness that each case can offer on its own terms.[15] Stake distinguishes between what he calls the instrumental case study and the intrinsic case study. For researchers using the former, the case is of secondary interest to the generalizations or theory that can be established. In the intrinsic case study, the research is "undertaken because one wants better understanding of this particular case."[16]

12.2.6 Distinguishing the Case Study

The case study, as we have described it here, is a distinct research design. Although the research literature may sometimes employ the term "qualitative case study,"

this does not mean that the case study research design is equivalent to or necessarily associated with the qualitative research design. Indeed, there is no necessity for qualitative research to adopt a case study design. And by the same token, case studies can be based almost exclusively on quantitative data (see section 12.3.2 in this chapter), or they may entail a theory-driven focus rather than the more inductively oriented approach frequently favored in much, though not all, qualitative research.

Similarly, the case study as a research design should also not be confused with case study *teaching*. True, both practices are defined by their focus on the case. But in the pedagogical context, the case materials can be deliberately altered to serve particular instructional purposes.[17] Such a practice is obviously antithetical to the principles and purposes of research. Nevertheless, case study research and case study teaching do both benefit from the compelling power of the robust and multifaceted character of the case focus.

12.3 STRATEGY: SINGLE OR MULTIPLE CASES?

Up to this point, we have focused on research examples of the single-case variety: urban vitality through the case of New York City, and popular Modernism through the case of Belo Horizonte, Brazil. Indeed, the single-case study can be highly compelling and—as with Jacobs's study—very influential. However, there are times when a researcher may want to consider a multiple-case design. On what basis does one make such a choice? And if one does choose a multiple-case design, how many cases are needed?

There is no quick and easy formula for making the choice *between* single- and multiple-case design, or about the *number* of cases necessary for a multiple-case design. But two principles are paramount, and both of them build on the special characteristics of the case study strategy identified in the previous chapter section: (1) the nature of the theoretical questions, or research questions, involved; and (2) the role of replication in testing or confirming the study's outcomes.

As single-case studies, both Jacobs's and Lara's work sought to investigate socio-physical phenomena involving multiple and highly complex factors. Each study dealt with issues from the scale of very broad cultural trends to the more intimate moments of sidewalk interaction and supervision of children (in Jacobs's case) or familial relationships represented in house plans (in Lara's study). From a theoretical point of view, it was more important for each of these researchers to uncover the very complex dynamics of one setting of interest than to limit the theoretical scope of the research by looking less deeply at more settings. From a practical

point of view, the level of complexity involved also suggested the virtue of a single-case design.

In other instances, a researcher may frame a theoretical question that is relatively narrower in scope, and in which identifiable factors of importance may vary from one case to another. In these circumstances, the multiple-case design might be advantageous. This raises the question: How many cases are enough? Here the answer is essentially the same as with the single-case design: the power of generalizability comes from the concept of replication, rather than the concept of sampling. Yin explains this quite well:

> [T]he decision to undertake multiple-case studies cannot be taken lightly. Here, *a major insight is to consider multiple cases as one would consider multiple experiments*—that is, to follow a "replication" logic. This is far different from a mistaken analogy in the past, which incorrectly considered multiple cases to be similar to the multiple respondents in a survey (or to multiple subjects *within* an experiment)—that is, to follow a "sampling" design.[18] (Emphases the author's)

This quotation highlights not only Yin's insistence on the significance of the replication logic, but also another important principle, namely that every case should serve a specific purpose within the overall scope of inquiry. To clarify what he means by this, Yin describes the distinction between *literal* and *theoretical* replication. A *literal* replication is a case study (or studies) that tests precisely the same outcomes, principles, or predictions established by the initial case study. In contrast, a *theoretical* replication is a case study that produces contrasting results but for predictable reasons.[19]

For example, if in the future Lara seeks to replicate his findings of the popular acceptance of Modernism in Belo Horizonte, he might seek to do so by conducting both literal and theoretical replications. In general, any other Brazilian case would be subject to the broad sociocultural trends of modernization, political history, and economy. However, a literal replication would also include conditions similar to those found in Belo Horizonte, specifically the building of an iconic exemplar of Modernism in that city, with the enthusiastic support of the city's political leadership. Without comparable conditions, another Brazilian case study city would constitute a theoretical replication. Similarly, if Lara were to study the popular acceptance of Modernism in another developing country, but without the overt support of Modernism by the nation's president, this would also constitute a theoretical replication.

With these considerations in mind, we will now consider two examples of multiple-case study designs.

12.3.1 A Multiple-Case Study: The Public Realm in College Towns

The goal of Anirban Adhya's research was to explore the nature of the public realm in everyday urban experience, specifically in terms of how public places are physically manifested, how they are understood, and how they are actually used. In doing so, Adhya sought to go beyond the formulations of contemporary theories and practices of urban design that conceive of publicness as "an undifferentiated and universally accessible place."[20]

To investigate this issue, Adhya chose to focus on the public realm in college towns, in part because they represent a distinct urban condition that features many of the qualities to which other cities aspire. Yet college towns face many of the same challenges in sustaining a viable public realm in the face of economic downturns, privatization, and competition from exurban developments.

Within this broad conceptual framework, Adhya concluded that he would study a limited set of case study towns in order to compare and contrast potentially similar and different conditions among them. Since there are no hard and fast rules about the number of cases to select, each researcher must assess the degree of depth versus comparative breadth that best suits the research question(s) he or she has posed. In this instance, Adhya chose to select two smaller university towns with campuses developed initially in the 19th century, and two state capital cities with campuses developed primarily in the mid-20th century. The four towns are: Ann Arbor, Michigan; Athens, Georgia; Tallahassee, Florida; and Lansing, Michigan. As indicated in Figure 12.13, Adhya decided to combine a *literal* and *theoretical* replication design. Thus, in theory the Ann Arbor/Athens and Tallahassee/Lansing pairs represented two *literal* replication pairs; in addition, he was able to test the outcomes of each replication pair against a pair of *theoretical* replications.

Another important feature of Adhya's research design is the elaboration of his primary research question about the nature of everyday experience of the public realm into a set of three subquestions concerning the experience of physical form,

College Town Types

Small—19th-Century Development	**Capital Cities—Mid-20th-Century Development**
1. Ann Arbor, Michigan	1. Tallahassee, Florida
2. Athens, Georgia	2. Lansing, Michigan

Figure 12.13 Multiple-case study design by Anirban Adhya.Courtesy of Anirban Adhya.

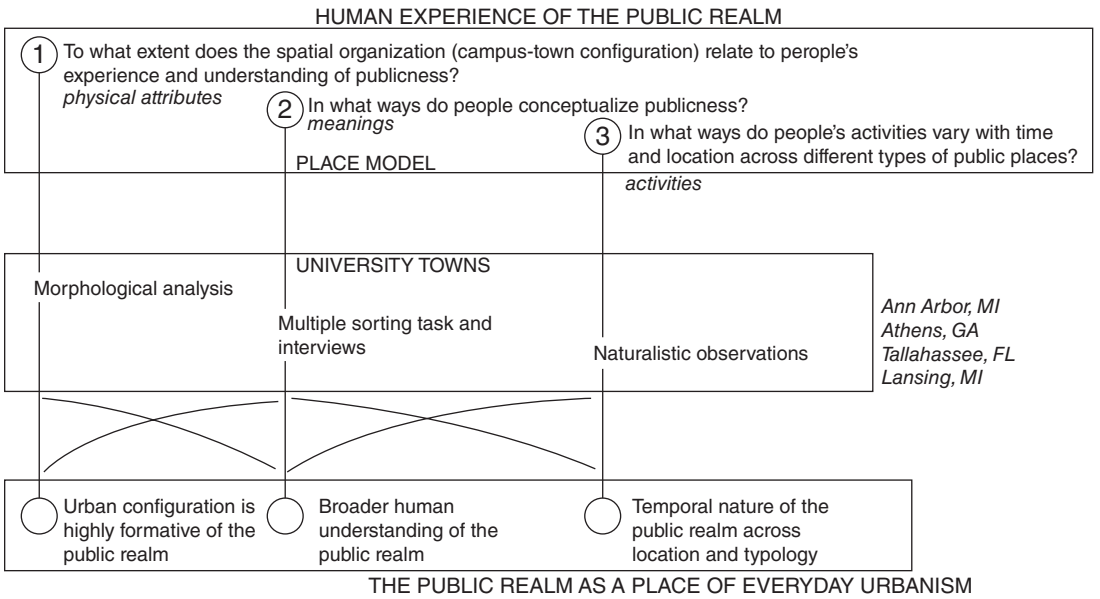


Figure 12.14 Research design for everyday experience of the public realm. Courtesy of Anirban Adhya.

qualities of meaning, and range of activities. For each of the three questions, he devised a distinctly appropriate tactic for data collection and analysis. This overall research design is represented in Figure 12.14.

In sum, this is an exceedingly robust study yielding many complementary and interrelated conclusions about the comparative experience of the public realm across the four cities. As it turns out, the initial categorization of the two literal and replication pairs was only partially confirmed. Although Ann Arbor and Athens are experienced in relatively similar ways (as are Tallahassee and Lansing), the four cities form more of a continuum of urban experience. This result was confirmed through multiple measures and analyses. Representative findings are presented in Figures 12.15 and 12.16.

12.3.2 A Multiple Case Study: Life-Cycle Assessment (LCA) of Low-Rise Office Buildings

Since one of the significant advantages of case study research is its capacity to investigate a setting or phenomenon embedded in its real-life context (see section 12.2.1), it is most commonly used in disciplines or specialty areas associated with

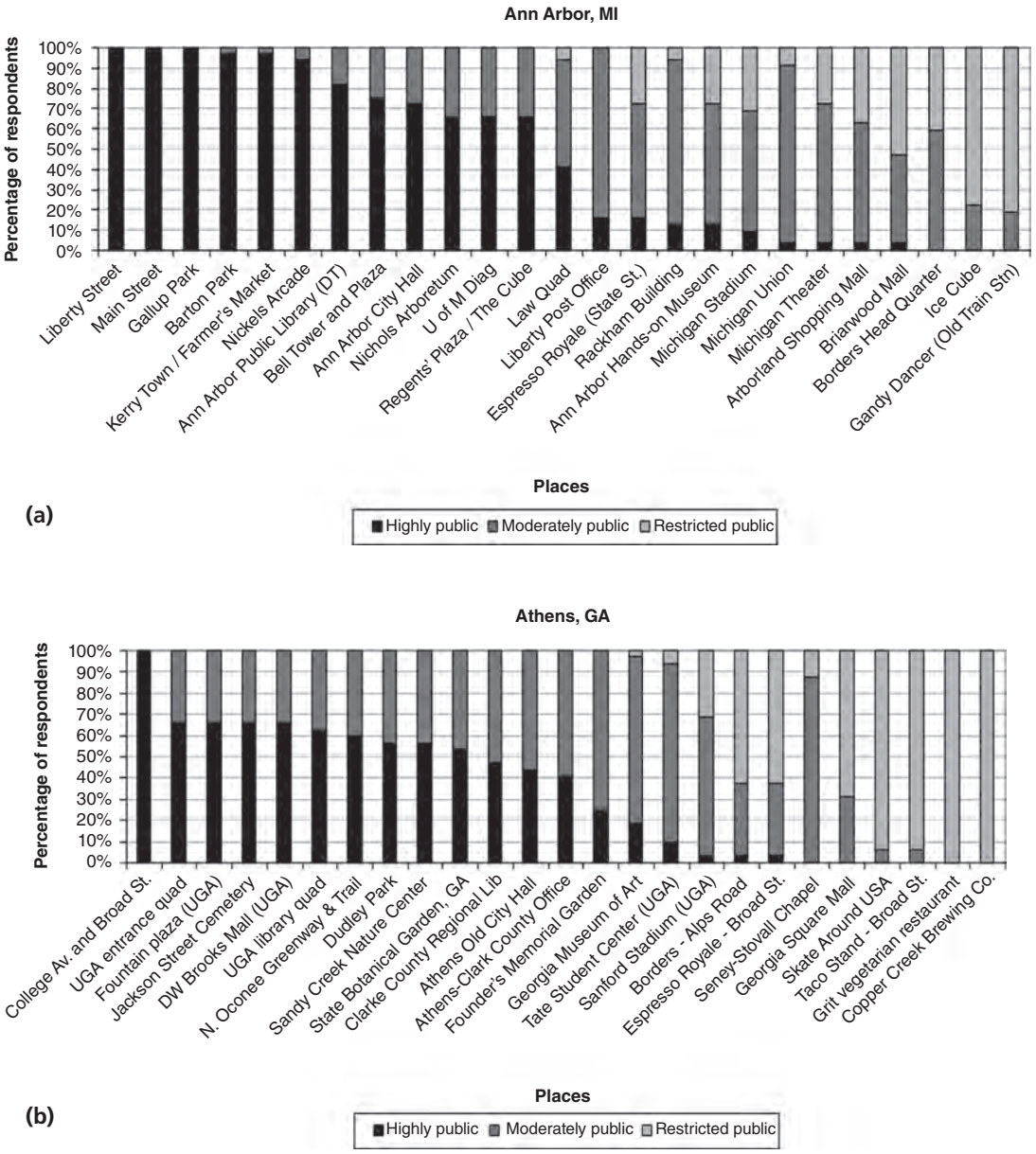


Figure 12.15 Distribution of responses for the directed sort in Ann Arbor, MI, and Athens, GA. Both Ann Arbor and to a lesser extent Athens have many places that are interpreted as highly or moderately public. Courtesy of Anirban Adhya.

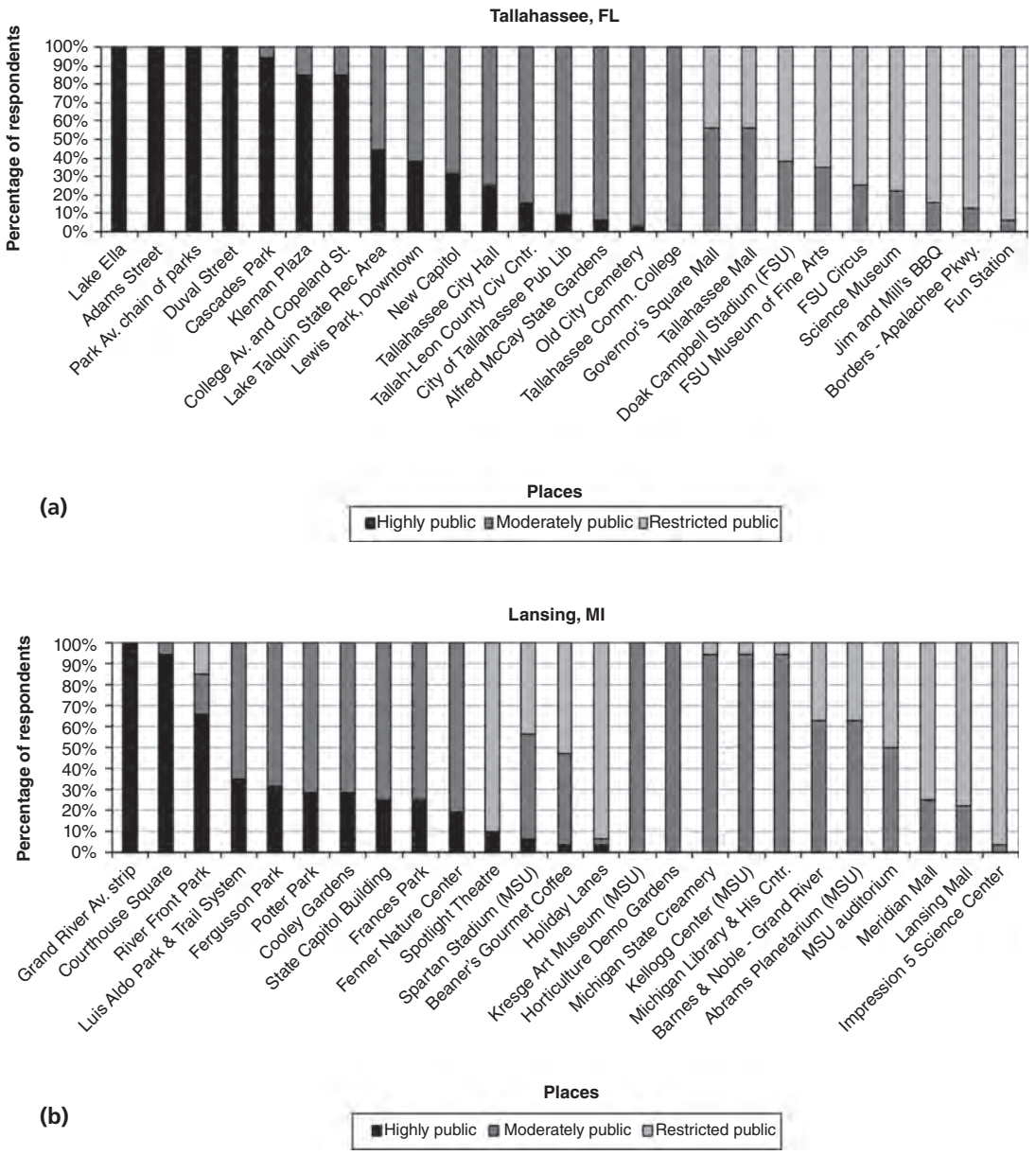


Figure 12.16 Distribution of responses for the directed sort in Tallahassee, FL, and Lansing, MI. Both Tallahassee and Lansing have relatively fewer places that are interpreted as highly public. Courtesy of Anirban Adhya.

sociocultural phenomena. And, although it is quite common for case study research to make use of both qualitative and quantitative data, it is relatively rare to find case studies that employ exclusively quantitative data.

Nevertheless, at least one relatively technical area of research that increasingly lends itself to case study research is that of sustainability. In this regard, Ashraf Ragheb's research using a life-cycle assessment (LCA) of low-rise office buildings suggests the potential for increased use of the case study strategy for more technical areas of environmental research. Ideally, LCA measures all inputs to a building and all outputs released to the environment.[21] Although earlier studies have employed LCA to assess individual buildings, they have tended to use fewer assessment criteria or limited life-cycle phases. Also, among the studies that have employed a more comprehensive focus, they have tended to apply LCA to residential buildings.

In this light, Ragheb's goal was to develop a fully comprehensive model that could be applied to the multiple case study of a set of commercial buildings. Ragheb chose three low-rise commercial office buildings as a set of literal replications. All three buildings share the following features: location in southeast Michigan, and therefore the same climatic conditions; relatively new construction, typical of a metropolitan suburban area; and low-rise configuration of one to four stories. In addition, for the purpose of actually accomplishing the research goal and measurement requirements, the owners and designers of each building were willing to participate in the study.

Because the range of LCA criteria were so comprehensive, it is beyond the context of this text to summarize the findings adequately. Although numerous comparative differences in impact were found among the three buildings, some overall generalizations could be derived. For example, the study found that the use phase of all buildings has the highest impact (over 90%) in the categories of energy consumption, global warming potential, and respiratory effects, whereas the manufacturing phase has the highest impact in the category of ozone depletion (87%).

BOX 12.3
Case Study: The Origins of Vancouverism

Robert Walsh's study of the emergence of "Vancouverism" critically examines the incremental process by which the distinctive urban architecture of Vancouver, British Columbia, developed through the combined contributions of several generations of local architects and planners. In

(*Continued*)

essence, Walsh's purpose was to answer the question: How did the development of urban form in Vancouver actually become "Vancouverism"?[a]

A term that first became popularized in 2004, "Vancouverism" is characterized by several critical features: separated residential point towers, a continuous street frontage of townhouses and low-rise buildings, a pedestrian streetscape with excellent street trees, a well-developed network of public waterfront parks, and a comprehensive system of protected natural views (see Figures 12.17 and 12.18). The net result is an environment in which a pedestrian-oriented planning ethos has been combined with moderately high development densities, producing a dynamic, commercially viable yet visually appealing urban environment.

Walsh's in-depth and multifaceted case study of Vancouverism draws on historical and archival material, interviews with key participants, extensive



Figure 12.17 Marina Crescent neighborhood at the Concord Pacific Place project, overlooking False Creek. Photograph by Robert Walsh.

[a] Robert Walsh, *The Origins of Vancouverism*. PhD dissertation, University of Michigan, Ann Arbor, MI, 2012.



Figure 12.18 Coal Harbor Mega Project, overlooking the Burrard Inlet and Vancouver Harbor. Photograph by Robert Walsh.

in situ observation, and detailed analysis of building designs and planning regulations. In particular, to illuminate the gradual evolution of Vancouver's urban architecture, Walsh undertook a critical assessment of many generally accepted but mistaken notions about how Vancouverism came to be, such as the popular misconception that it was recently imported from Asia. Although there has been a recent influx of immigration and capital from Asia that has helped to fuel the rapid development of Vancouver, the building forms themselves were not imported from Asia. In the process of unraveling the Asian import myth and other popular myths, a new picture began to emerge.

Indeed, Walsh discovered that the development of Vancouverism can be traced through the work of local architects and planners, struggling over an extended period of time to find solutions that made sense within the particular context of Vancouver. This context includes a combination of influential local factors: economic, climactic, geographic, cultural, political,

(*Continued*)

and historical. In choosing to focus on these local factors instead of relying upon established standard solutions, the planners and architects working in Vancouver gradually developed a different form of urbanism better adapted to the unique needs and conditions found in their city.

Perhaps the most significant outcome is that Walsh's research effectively challenges the widely held view that Vancouverism represents a general typology useful in other urban contexts. Instead, as a locally developed response to an exceptional context, Vancouverism should probably not be expected to work well elsewhere. Instead of continuing to copy Vancouverism in foreign cities, planners and designers might actually learn more from the example of Vancouver by appreciating more fully the importance of cultivating their own local urban ideals, and relying upon the local knowledge base. In this way the example of Vancouverism has the potential to point the way to a vast new range of unique local urbanisms each well suited to its setting, each relying upon local creativity, and each a reflection of local values of the community.

12.4 CASE STUDIES: STRENGTHS AND WEAKNESSES

Many of the strengths of the case study research design flow naturally from the defining characteristics already identified in section 12.2. To be specific, the first four items listed in Figure 12.19 directly mirror those earlier section headings. However, those acknowledged advantages also give rise to the potential for corresponding weaknesses. For instance, the embeddedness of the case study in its context can lead to such an expansion of scope that the study itself becomes unwieldy. And although the case study can uncover and explain causal links, this understanding of causality is likely to be more variegated and multifaceted than the attribution of causality to a "treatment" in an experimental design. Similarly, while the incorporation of multiple data sources can provide confirmation of findings across different data sources, establishing the coherence of the study as a whole becomes more challenging. Not least, although meeting the standard of generalizing to theory—rather than from sample to population—represents an important strength of the case study, confirmation of the theory still requires replication in other case studies.

Taken together, these strengths and weaknesses lead to the more general assessment listed under item 5 in Figure 12.19. Indeed, the depth, complexity, and multifaceted quality of the case study contribute to its robust capacity as a research design. Examples such as Jacobs's study of city life reinforce the point that a case

Strengths	Weaknesses
1. Focus on the embeddedness of the case in its context	1. Potential for overcomplication
2. Capacity to explain causal links	2. "Causality" likely to be multi-faceted and complex
3. Richness of multiple data sources	3. Challenge of integrating many data sources in coherent way
4. Ability to generalize to theory	4. Replication required in other cases
5. Compelling and convincing when done well	5. Difficult to do well; fewer established rules and procedures than other research designs

Figure 12.19 Case study strengths and weaknesses.

study, when done well, can be a particularly compelling form of research. However, there are fewer established rules and procedures for designing and conducting case study research. Almost by definition, the case study is conceived in terms of case-particular considerations, and great latitude is afforded the researcher in devising the overall research design and selecting a particular combination of tactics. This contrasts quite sharply with the much more prescribed components of the experimental strategy, or the measurement of clearly specified variables in correlation research. Although just "following procedures" in experimental or correlational designs by no means guarantees quality, it is even more true that the case study requires the researcher to carefully think through both the overall framework and the details of the research design.

12.5 COMBINED STRATEGIES: INTEGRATING MULTIPLE RESEARCH DESIGNS

Increasingly, researchers in many fields, including architecture, are advocating a more integrative approach to research whereby multiple methods from diverse traditions are incorporated in one study. Because each typical research strategy brings with it particular strengths and weaknesses (as we have noted in the previous chapters), many researchers believe that combining methods provides appropriate checks against the weak points in each, while simultaneously enabling the benefits to complement each other.

Indeed, over the past two decades, there has been an explosion of interest and active development of the mixed methods concept in a variety of disciplines. In an

article in the newly initiated *Journal of Mixed Methods Research*, author Jennifer Greene argues that mixed methods advocates are particularly active in fields that involve "a dynamic interplay with creative practice in highly practical fields."[22] Greene and the founding editors of the journal identify a number of fields as active areas of mixed methods research, including architecture, educational psychology, nursing and health care, management, sustainability, anthropological demography, and development economics.[23] Although readers with design backgrounds might question the extent to which "creative practice" is embodied in some of these fields, Greene's argument is that each of these fields depends upon a dynamic relationship between "thinking/knowing and acting/doing."[24]

A potentially confusing aspect of the discussion, however, is that the many authors writing on mixed methods often do so at very different levels of the research process, such as specialty topic areas, paradigms, schools of thought, research design, or the level of tactics in data collection. For example, in discussing the apparent paradigmatic dichotomy in architectural research (framed as myth vs. science), Robinson[25] offers a number of examples of integrative research described primarily in terms of specialty areas. (See Chapter 3 for a detailed discussion of this article.) In particular, she describes Geoffrey Broadbent's work as combining design methods and human behavior;[26] Susan Ubbelohde's work as an integration of technology, history, and human behavior;[27] and Juan Paul Bonta's research as linking mathematics, art, and social science.[28]

More recently, some authors have claimed that the mixed methods perspective constitutes its own "paradigm." For example, Tashakkori and Teddlie (editors of the *Handbook of Mixed Methods*) argue that mixed methods research is best conceived under the rubric of pragmatism.[29] (In Chapter 3 we defined pragmatism not as a distinct system of inquiry, but as a school of thought.) However, even among authors who situate mixed methods within the framework of pragmatism, there is some disagreement as to whether to emphasize pragmatism as deeply rooted in the philosophical tradition of Dewey or as a more generically pragmatic sensibility. In contrast, educational psychologist Donna Mertens has situated mixed methods as more appropriately framed by the transformative-emancipatory tradition derived from critical theory—another tradition earlier identified as a school of thought in Chapter 3.[30]

Yet other authors have argued that, rather than conceiving mixed methods as a distinct paradigm or school of thought, it is more appropriate to recognize the diversity of paradigms *within* mixed methods. To that end, author Gitte Harrits analyzes two mixed methods studies: one in the field of comparative politics by E. S. Lieberman, in which the goal was to strengthen causal inference; the second, a study by French sociologist Pierre Bourdieu on the concept of "habitus" (defined as systematic patterns in actions and events), in which he employed survey data in

combination with "interpretive analysis of texts, pictures, and interviews."[31] As a result of her in-depth analyses, Harrits identifies two distinct sets of underlying epistemological and ontological positions. Whereas Lieberman's study embodies a critical realist stance using both quantitative and qualitative data to confirm causal mechanisms within the "same" understanding of "reality," Bourdieu's study assumes that society is understood as both a "system" and a "lifeworld," thereby representing two realities that are necessarily not "translatable."

Despite the competing arguments about the predominance, or lack thereof, of a particular paradigm or school of thought within mixed methods research, most authors tend to describe the actual mixing of methods within a given study as a combination of quantitative and qualitative data techniques.[32] Indeed, over the past decade, many advocates of mixed methods research have developed complex typologies of research designs specifying options for sequencing QUAN and QUAL components of a research project. Readers of this book who choose to adopt a mixed methods approach will certainly benefit from clarifying how they might employ both quantitative and qualitative sources and analyses; we recommend the sources referenced above as useful guides.

Nevertheless, we would argue that emphasizing the level of tactics (quantitative or qualitative) in mixed methods research may obscure the broader issues of research design that may be central to the complex fields of architecture and design research. Given that environmental design research necessarily addresses the complicated dynamics of physical form/settings, purposive actions, and interpretations of meaning over time, many studies are likely to encompass a broader range of research designs than in other fields or disciplines. Indeed, the research design types presented in the previous six chapters of this book are testament to this range and diversity in our fields.

For this reason, we will focus the following discussion of mixed methods primarily at the level of strategy, or research design. For instance, how might a researcher go about combining the historical strategy with simulation, or the correlational with the qualitative? Although there are no quick formulas for developing a research design in the first place, much less a combined strategy, Creswell does offer three general models that are suitable for our consideration: (1) the two-phase approach; (2) the dominant–less dominant design; and (3) the mixed methodology design.[33] In the following chapter sections we will discuss each model in turn, along with appropriate examples of actual research studies.

12.5.1 A Combined Strategy: A Two-Phase Design

As the term itself suggests, a two-phase research design involves combining two or more strategies in a sequence of distinct phases. The advantage of such an approach

is that the particular procedures and standards associated with each strategy can be presented fully and distinctly. A possible disadvantage is the potential for a perceived lack of connection or coherence if the strategies are not conceptually well linked.

One example that well represents the potential advantages of a two-phase mixed method design is Wall et al.'s study of commuting mode choice among staff and students at a university in England's midland region.[34] The research design was conceived by the authors such that phase 1 focused on respondents' intentions, while phase 2 focused on actual behavior; in that sense, the two phases were complementary. In phase 1, a quantitatively based survey questionnaire was employed; and based on the data analyses for phase 1, a purposive sample of participants (representing typological responses) was selected for in-depth interviews. For this second phase, the authors' goal was to enable participants to explain the various influences on their travel behavior in their own terms.

Overall, the authors found that although respondents' intentions for limiting car use were strongly influenced by personal-normative motives (phase 1 results), their actual behavior was significantly influenced by their perceived behavioral control, that is, the lack of practical options for avoiding the use of a car. On a policy level, this study led to the implementation of many environmental interventions that could lead to increasing perceived control over choice of travel mode.

In methodological terms, the authors conclude that since the study aimed to contribute to the development of theory (explaining how participants' mental constructs interacted), the mixed methods design demonstrated that the interview results were consistent with the statistical results of phase 1. Moreover, the authors note that the research design embodies the description of abductive reasoning "that moves back and forth between induction and deduction—first converting observations into theories and then assessing those theories through action." This, the authors claim, demonstrates the value of a research design that is both "theory-led and data-led."[35]

In contrast to the travel mode study, it is also possible to employ a two-phase research design without adopting an overall case study design. For example, Ahrentzen and Groat conducted an extensive study of the status and viewpoints of women faculty in architecture that utilized a correlational design as the first phase and a qualitative design for the second phase. In the first phase of the research,[36] the authors employed a survey questionnaire that was sent to all faculty women in architecture, as well as a shorter survey that was sent to architecture program chairs. Although the bulk of their report from this phase utilized descriptive statistics, they also highlighted the relationships between the faculty women's and the chairs' perceptions of women's status in their schools.

In the second phase of the research, Groat and Ahrentzen[37] adopted a qualitative research strategy. They made use of the faculty women's perceptions, especially their responses to two open-ended questions, as a basis for developing the interview protocol for in-depth phone interviews with approximately 40 faculty women respondents. In this phase, in contrast to the survey phase, there was no intention to achieve a random sample of respondents; rather, "the goal was to maximize the variety and range of perspectives."[38] As a consequence, the sample of respondents was more heavily weighted to tenured women, precisely the group of women who are more likely to exert some influence within the academy. The interviews themselves entailed three broad themes: attractions to architecture as a career, career experiences, and visions of architectural education. The outcome of this second phase was an analysis of the extent to which the faculty women's perspectives mirrored aspects of the recommendations of the Carnegie Foundation study of architectural education.[39]

12.5.2 A Combined Strategy: A Dominant–Less Dominant Design

As its name suggests, the dominant–less dominant design entails the insertion of one type of research design within the framework of a distinctly different research design. The advantage of this design is that it offers the potential of maintaining the overall coherence of the study as it is vested in the dominant research design. The less dominant design is then used to provide greater depth and/or validity concerning a particular aspect of the study. The consequent disadvantage is that the full and potentially complementary strengths of the less dominant design will not be fully realized.

A useful example of this dominant–less dominant design is Joongsub Kim's study of New Urbanism in Kentlands[40] and a comparable conventional suburban design, described in considerable detail in Chapter 8. Earlier, we characterized Kim's study as a causal comparative design, under the more general category of the correlational strategy. This indeed is the dominant strategy of the study. The primary means of data collection was a detailed survey questionnaire designed to elicit from residents of the two neighborhoods their perceived sense of community. The questionnaire asked the respondents to evaluate the extent to which they believed that various physical features affected their responses to four different indicators of community: sense of attachment, pedestrianism, social interaction, and sense of place identity. As is typical of the correlational strategy, various statistical calculations were used to assess the relationships among multifaceted variables measured in the study.

However, included in Kim's study is a secondary element: in-depth interviews with over 100 residents, which generally conform to a qualitative research strategy.

Kim had several reasons to incorporate such interviews as a less dominant component in his study. First, he wished to explore in a more open-ended fashion and in greater depth some of the themes from the survey questionnaire: for instance, narrative vignettes of the kind of social interactions they had with their neighbors, and how such interactions affected their sense of community. A second reason Kim chose to include the qualitative interviews is that he has no way to establish the comparability of the two different neighborhood groups (a weakness of his causal comparative design). Is the reason that Kentlands residents evaluate their sense of community more highly because the people who chose to move there are more community-minded in the first place? Through the interviews, Kim was able to establish that many—though not all—residents had moved to Kentlands without any particular intention about social interaction, pedestrianism, or community sensibilities, but once in residence their habits and inclinations had changed.[41] Thus, as a consequence of these interviews Kim was able to address directly a potential threat to the validity and interpretation of his study.

12.5.3 A Combined Strategy: Mixed-Methodology Design

The mixed-methodology design represents the most complete level of integration among two or more research designs. In this model, the researcher would conduct aspects of both strategies in roughly comparable sequences, and with approximately equal degrees of emphasis. The advantage of such an approach is that presumably the strengths of each research design will complement each other, while the weaknesses of each design will be substantially offset. However, the mixed methodology may well require a level of sophistication in multiple research designs that is not always common for people trained in a very specific research tradition. Moreover, some "purists" may find the combination of research designs too unconventional and therefore suspect.

A good example of a mixed-methodology design is Lara's study of popular Modernism in Brazil, a study we have already described in considerable detail as a case study.[42] In fact, it is clearly both, as we noted earlier in this chapter; because the case study strategy typically makes use of multiple sources of evidence, it may frequently entail the combination of not just data collection tactics but also distinct research designs. Indeed, in Lara's study the combination of historical and qualitative designs is virtually seamless (see Figure 12.20). Within the historical design, he has included extensive archival research on the portrayal of Modernism in popular media, archival documentation of Modernist house plans in the city offices of Belo Horizonte, an artifactual inventory of all the houses in two multiblock areas of the city, and a detailed stylistic analysis of each house facade within the neighborhood

Strategies	Tactics: Data Sources
Interpretive-Historical	Housing inventory of 2 neighborhoods Stylistic analysis of all houses Archival documentation of houses in city records Verbal / visual analyses of media representations
Qualitative	In-depth, open-ended interviews with original residents Configurational analyses of representative house plans Computer-based spatial analyses of house plans

Figure 12.20 Diagram of combined strategies of Fernando Lara.

inventories. Also note that Lara has, even within this historical strategy, incorporated not only verbal and visual data (e.g., the media analyses) but also more quantitative data (e.g., the housing inventories).

Second, Lara simultaneously interweaves a qualitative research design within his study. Included within this component are open-ended interviews with 21 original residents of popular Modernist homes in Belo Horizonte; and detailed spatial analyses of the house plans, linked with the residents' commentaries on the social dynamics of their family life. Just as Lara incorporates both quantitative and qualitative data and analyses in the historical component, he also incorporates both quantitative and qualitative elements in this segment of his study. For example, the floor plan analyses depend on quantitative measurements as well as social history provided by the residents; also, verbal, qualitative analyses of the floor plans are complemented by computer analyses of "spatial depth." In sum, Lara's deft and complementary use of two research strategies, as well as both quantitative and qualitative tactics within each, makes this study a very telling example of the mixed-methodology design.

12.6 COMBINED STRATEGIES: STRENGTHS AND WEAKNESSES

Figure 12.21 briefly summarizes the relative strengths and weaknesses of the three models of combined strategies that were presented in each of the separate chapter segments. Taken together, the chart suggests that while there is much to be gained by integrating different research designs, the researcher may also find that combining strategies requires a higher level of sophistication in research methodology than would be expected if he or she were to use a more conventional approach. It is still the case that in many academic disciplines, and certainly in architecture, particular research designs are often taken for granted as the preferred method for research in

Model of Combination	Strengths	Weaknesses
1. Two-phase	Each strategy can be presented fully and distinctly	Potential lack of connection and coherence
2. Dominant – less dominant	Potential for maintaining coherence through emphasis on dominant design Less dominant design can provide depth and validity	Complementary strengths of less dominant design not fully realized
3. Mixed methodology	Potential to maximize strengths and minimize weaknesses of each design	Need for level of sophistication in multiple research design Mixed methodology too unconventional for some purists

Figure 12.21 Strengths and weaknesses of combined strategies. Adapted from J. Creswell, *Research Design: Qualitative & Quantitative Approaches* (Thousand Oaks, CA: Sage, 1994), 177–178. Reprinted by permission of SAGE Publications.

particular topic areas. This means that many researchers have been exposed to and trained primarily in one or perhaps two research strategies; as a consequence, it may require considerable effort to go beyond and augment the preferred strategy with a suitably complementary strategy.

Even if the researcher is knowledgeable about multiple research strategies, there still remains the challenge of how to combine strategies in an effective and coherent way. To be specific, the two-phase design may put the overall coherence of the study at risk in the effort to highlight the distinctly different qualities of separate research strategies. However, the dominant–less dominant design tends to privilege the coherence of the study by placing the less dominant strategy in a secondary role, thus compromising the potential strengths of that less dominant strategy. Finally, the mixed-methodology model tends to present the greatest challenge for the researcher in reconciling and integrating two (or more) disparate strategies. Yet, if done well, it may yield the greatest payoff, as the complementary strengths of the combined strategies may be most fully realized.

Despite the assorted pitfalls and challenges, it is nevertheless our contention that combined strategies research represents an enormous opportunity for architectural research. By definition, architecture is a multidisciplinary professional field.

Yet, to date, much architectural research has continued to be conducted within the confines of subdisciplinary topic areas, such as environmental technology or architectural history.

Certainly, there will always be a valuable role for research efforts in these and other traditionally defined areas, but likewise there are many other important topic areas that defy easy categorization. Should research on environmental comfort or on energy conservation habits of building users be considered environmental technology research or behavioral research? If it is both, then shouldn't some combination of the research designs typically used in each research tradition be combined so as to investigate the phenomenon or setting in a more effective and multifaceted way?

Because there are, at this point, fewer established rules and procedures for designing combined research strategies, the researcher must exercise more care and build on a greater range of knowledge in research methodologies. But, despite this higher level of challenge, we believe that architectural research that combines strategies represents an important and necessary frontier in our field.

NOTES

1. Jane Jacobs, *The Life and Death of Great American Cities* (New York: Random House, 1961), 13, 23, 25–26.
2. Fernando Lara, *The Rise of Popular Modernist Architecture in Brazil* (Gainesville: University Press of Florida, 2008).
3. Robert K. Yin, *Case Study Research: Design and Methods*, 4th ed. (Thousand Oaks, CA: Sage, 2009), 18.
4. Ibid., 176.
5. Jacobs, 23.
6. Ibid., 25.
7. Ibid., 23–24.
8. Yin, 35–36.
9. Ibid., 36.
10. Jacobs, 25.
11. Ibid., 25.
12. Lara, 1–2.
13. Ibid.
14. Yin, 44.
15. Robert E. Stake, "Case Studies," in Norman Denzin and Yvonna Lincoln (eds.), *Strategies of Qualitative Inquiry* (Thousand Oaks, CA: Sage, 1998), 86–109.
16. Ibid., 88.
17. Yin, 14.

18. Ibid., 53–54.
19. Ibid., 54.
20. Anirban Adhya, *The Public Realm as a Place of Everyday Urbanism: Learning from Four College Towns.* PhD dissertation, University of Michigan, Ann Arbor, MI, 2008.
21. Ashraf Ragheb, *Towards Environmental Profiling for Office Buildings Using Life Cycle Assessment (LCA).* PhD dissertation, University of Michigan, Ann Arbor, MI, 2011.
22. Jennifer Greene, "Is Mixed Methods Social Inquiry a Distinctive Methodology?" *Journal of Mixed Methods Research* 2(1) (2008): 8.
23. Abbas Tashakkori and John W. Creswell, "Mixed Methodology across Disciplines," *Journal of Mixed Methods Research* 2(1) (2008): 5.
24. Greene, 8.
25. Julia Robinson, "Architectural Research: Incorporating Myth and Science," *Journal of Architectural Education* 44(1) (1990): 20–32.
26. Geoffrey Broadbent, *Design in Architecture* (Chichester, UK: John Wiley & Sons, 1973).
27. Susan Ubbelohde, "Oak Alley: The Heavy Mass Plantation House," *Eleventh Annual Passive Solar Conference Proceedings,* American Solar Energy Society, 1986.
28. Juan Paul Bonta, *Architecture and Its Interpretation: A Study of Expressive Systems in Architecture* (New York: Rizzoli, 1979).
29. Abbas Tashakkori and Charles Teddlie, "Major Issues and Controversies in the Use of Mixed Methods in the Social and Behavioral Sciences," in Abbas Tashakkori and Charles Teddlie (eds.), *The Handbook of Mixed Methods in Social & Behavioral Research* (Thousand Oaks, CA: Sage, 2003), 20.
30. Donna Mertens, "Mixed Methods and the Politics of Human Research: The Transformative-Emancipatory Perspective," in Abbas Tashakkori and Charles Teddlie (eds.), *The Handbook of Mixed Methods in Social & Behavioral Research* (Thousand Oaks, CA: Sage, 2003), 159.
31. Gitte Harrits, "More than Method? A Discussion of Paradigm Differences within Mixed Methods Research," *Journal of Mixed Methods Research* 5(2) (2011): 150–166; Evan Lieberman, "Nested Analysis as a Mixed-Method Strategy for Comparative Research," *American Political Science Review* 99(3) (2005): 435–452; Pierre Bourdieu, *Distinction: A Social Critique of the Judgement of Taste* (London: Routledge, 1984).
32. Abbas Tashakkori and Charles Teddlie (eds.), *The Handbook of Mixed Methods in Social & Behavioral Research* (Thousand Oaks, CA: Sage, 2003); John W. Creswell, *Research Design: Qualitative & Quantitative Approaches*, 3rd ed. (Thousand Oaks, CA: Sage, 2009); Charles Teddlie and Abbas Tashakkori, *Foundations of Mixed Methods Research: Integrating Quantitative and Qualitative Approaches in the Social and Behavioral Sciences* (Los Angeles: Sage, 2009).
33. Creswell, 206–207.
34. Rob Wall, Patrick Devine-Wright, and Greig Mill, "Interactions between Perceived Behavioral Control and Personal-Normative Motives: Qualitative and Quantitative Evidence from a Study of Commuting-Mode Choice," *Journal of Mixed Methods Research* 2(1) (2008): 64.

35. Ibid., 83.
36. Sherry Ahrentzen and Linda Groat, *Status of Faculty Women in Architecture Schools* (Washington, DC: Association of Collegiate Schools of Architecture, 1990); Sherry Ahrentzen and Linda Groat, "Rethinking Architectural Education: Patriarchal Conventions & Alternative Vision from the Perspectives of Women Faculty," *Journal of Architecture and Planning Research* 9(2) (1992): 95–111.
37. Linda Groat and Sherry Ahrentzen, "Voices for Change in Architectural Education: Seven Facets of Transformation from the Perspectives of Faculty Women," *Journal of Architectural Education*, 50(4) (1997): 271–285.
38. Ibid., 273.
39. Ernest Boyer and Lee Mitgang, *Building Community: A New Future for Architecture Education and Practice* (Princeton, NJ: Carnegie Foundation for the Advancement of Teaching, 1996).
40. Joongsub Kim, "Perceiving and Valuing Sense of Community in a New Urbanist Development: A Case Study of Kentlands," *Journal of Urban Design*, 12(2) (2007): 203–230.
41. Ibid.
42. Lara.

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